

Anthony Zheng

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EDUCATION

University of Minnesota

Bachelor of Science in Computer Science - 3.8 GPA (Dean's List 5 Semesters)

Minneapolis, MN

2022 - 2026

EXPERIENCE

Software Engineer

March 2026 - Present

Visa Inc.

Austin, TX

- Developed the ingestion and linking engine for a GraphRAG service using **Apache AGE** and vector extensions on **PostgreSQL**; built the core logic to map code-to-test relationships via contextual embeddings.
- Developed the analysis engine for a LangChain-powered Test Execution Agent, implementing deterministic log parsing and batch processing capable of scaling to handle massive test reports containing **up to 5,000+ test cases**.
- Reduced test failure analysis and triage time by 75% (saving an estimated **10+ hours/week**) by extracting logs from LogSimplify to identify root-cause errors and dynamically update executions and cycles to **Zephyr**.

Software Developer Intern

May 2025 – August 2025

Visa Inc.

Austin, TX

- Built a **NextJS** application leveraging **agentic commerce** to enable purchases through Visa's payment network
- Integrated a **Langchain** agent, connecting users to web search and **MCP servers**
- Deployed application using **Cloudflare Workers** for internal use and testing

Software Developer Intern

June 2024 – August 2024

Aspen Technology

Medina, MN

- Implemented a tree map component to Voyager using TypeScript, allowing **thousands** of customers to configure and display data in different ways
- Processed data from Aspen Tech's proprietary **DBMS** as well as from **REST APIs** to be compatible with **DevExtreme** components
- Collaborated with the team to triage and resolve tickets, addressing issues from UI inconsistencies to database errors

PROJECTS

Money Manager | *Next.js, PostgreSQL (Neon Serverless), Docker, GCP*

- Engineered a finance platform using **Next.js, React, and TypeScript**, integrating NextAuth.js for **Google OAuth**, and leveraged **Google Cloud Scheduler** to run cron jobs for recurring transactions
- Reduced server-to-DB network latency by **51%** by co-locating **Google Cloud Run** instance in us-east4 with AWS-hosted Neon DB (us-east-1) to eliminate cross-cloud network overhead.
- Optimized database query performance by migrating indexes to a composite **B-Tree** structure (userId, tab, date), achieving a **37%** reduction in query latency.
- Configured an automated **Google Cloud Build** pipeline to build and push production Docker images to **Artifact Registry**, utilizing Next.js standalone outputs to reduce image size by **85%**
- Architected an event-driven transaction ingestion pipeline utilizing iOS Shortcuts, **GCP Cloud Scheduler**, and secure **IMAP** polling, leveraging **Gemini 2.5 Flash** to semantically categorize raw data and automate **80%** of manual entries.
- Engineered a DB-first merchant name matching algorithm that pre-screens normalized vendor names, bypassing LLM calls for repeat purchases and reducing **Gemini API** requests by **75%+**.

League of Legends Data Tracker (GSB) | *Angular, Python, Flask, Git*

- Utilized Riot Games API to get data on **millions** of players and their match performances
- Developed a front-end UI using Angular to display the data concisely, which allows users to improve their gameplay

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, Java, C/C++, HTML/CSS, SQL

Frameworks: React, NextJS, Node.js, Flask, Angular, Express

Technologies: Git, Docker, VSCode, IntelliJ, MongoDB, Jira, OpenShift